NFAI Submission Content

https://www.sprind.org/taten/challenges/submissions/next-frontier-ai

PROJECT TITLE 50

Reviewer focus: A crisp title that signals the frontier and the system, not just a codename.

DAEDALUS: Compositional Intelligence

SHORT DESCRIPTION 493/500

Reviewer focus: One paragraph: problem, core innovation, why now, expected outcome.

The brain runs general intelligence on ~20W, about three bananas; a DGX B200 draws ~14kW, roughly 2,100 bananas, for one narrow model. That 700:1 gap is the clue: nature's intelligence is shaped by information per unit energy, by organised structure rather than scale. Six components build a 1B model with any-to-any multimodality including action, world-model planning, persistent memory and audit traces, running on phones, robots and vehicles. We target frontier capability at ~1/100th the compute

FRONTIER DIMENSION 482/500

Reviewer focus: Which new frontier does the idea target? Examples from form: model architecture, data/training paradigm, multi-agent orchestration, safety/alignment, domain-specific AI application.

Our compositional research programme unlocks a new architecture and training paradigm with safety/alignment controls. Organised structure and function are the missing ingredients for leapfrog technology. Capability-per-Watt is the test; research components are designed to yield high information even in failure. Resource-rational AI that allocates compute by information value, trains beyond next-token, holds persistent memory, simulates worlds in latent space, and self-improves under audit.

CORE IDEA & ARCHITECTURE 3,000

Briefly describe the end-to-end system: main components, data flow, and how they interact.

Our compositional research pushes AI from a single-minded neo-cortical structure (transformers) trained on a biologically incompatible objective (next-token prediction with end-to-end backprop), to a new frontier complex-system mind that respects energy, structure, function, and scale.

The system

DAEDALUS is any-to-any multi-modal, with action as a first-class output, built to make decisions, not just predictions: it forecasts consequences, weighs options, and acts. Small enough to run on a phone, a robot, or in a vehicle, it has world-model planning, persistent memory (shareable across instances), and governance controls (normative checks). Organised around explicit information gain per watt, it is built from six research components (a seventh, learning beyond backprop, appears under Technical Novelty):

C1. Better learning signals. Within a context window W, at token position t we train three objectives jointly: backward prediction of preceding tokens (t−n to t−1), fill-in of a randomly masked subset across W, and forward prediction of following tokens (t+1 to t+n). This improves causal understanding and collapses the 'lost in the middle' curve behind NTP-model forgetting.

C2. World modelling. The model predicts the next state of the world across vision, sound, sensors and language. Able to forecast how the world responds to an action, it plans, simulating candidate actions in latent space and choosing the best outcome before acting.

C3. Explicit memory and recursive latent reasoning. The model has dedicated read/write memory pools (Neural Turing Machines) for larger context. It reasons recursively in a compact latent space rather than in tokens or logits, end-to-end learnable and far cheaper at every step.

C4. Compute-aware attention. The model learns where to spend compute by predicting which inputs carry the most information. Capability-per-Watt becomes something we optimise for, not measure afterwards.

C5. Self-improvement loop. A model generates training data for the next, which trains an improved model. Each cycle is grounded in real interaction data and gated by C6's audit; an anti-fragile loop, not model collapse.

C6. Built-in norms. Moral and regulatory requirements become explicit, checkable rules constraining inputs and outputs. On a breach, the system explains why and alerts the user. Compliance is an architectural property, not a wrapper bolted on afterwards.

How the components interact

The six compose rather than add. C1's past-and-future objectives create the temporal signal C4 needs to learn where compute is worth it. C3 gives the stack a movable state and recursive reasoning, and C2 puts planning in a multi-modal world model. C5 keeps it fed from reality; C6 gates what it may learn. Each coupling raises information per watt: how a 1B model reaches 100B-class capability. Stage 1 measures these as scaling laws over modality count, prediction horizon, memory size, and routing granularity, and falsifies them if they fail.

TECHNICAL NOVELTY 2,000

Which cutting-edge techniques (cite the most recent papers or pre-prints) are you combining, and why do they give a measurable advantage over the current state-of-the-art? Which capability gap in current AI systems, unlikely to be addressed through the continued development of existing systems, does this submission address?

We target seven core shortcomings of transformers. Not all will succeed; we pursue the highest-information bets, where even a failure advances the field.

C1. Train on more than the next word. One pass predicts past, present and future, enforcing causality and collapsing the lost-in-the-middle curve, on a brain-motivated objective [1,2,3]. Parallel decoding for free.

C2. World modelling for planning. Predicts the next multimodal world state in latent space [4,5,6], not text about it, so it plans before acting. Grounded in GATE/TALI and Omniverse [7].

C3. Memory and recursive latent reasoning. Read/write slot memory [8] and recurrent reasoning on a low-dimensional latent manifold [9,10,11], surfacing to tokens only. MNEME [12]: a small model reaches near-100% long-context recall versus a baseline near chance; recursion is compositional depth, not generic compute.

C4. Compute as a trained objective. Unlike post-hoc sparsity [13,14], the model routes compute by predicted information value, making Capability-per-Watt a training target. Routing baselines already match dense N^2 attention on toy tasks at matched compute.

C5. Audited self-improvement. The model generates its next training data and annotates decisions, gated by C6: anti-fragile, not collapse-prone. Our multi-agent error-correction system scores rho=0.74 vs experts.

C6. Governance by construction. Not a wrapper: norms compile into every component, with audit traces meeting EU AI Act logs [16,17], co-designed by a P7001 author.

C7. Learning beyond backprop. Evolution optimises what backprop cannot: non-differentiable parts, joint arch/hyperparameter search. Our einspace work [15] composes architectures from primitives, already ~75% of an AdamW model.

Today's frontier scales one architecture (the transformer), one objective (next-word) and one optimiser; memory, planning, governance and efficiency get bolted on. We build the integrated, resource-rational architecture that scaling cannot reach [18].

TECHNICAL NOVELTY CITATION 1,000

[1] Berglund et al. 2023, Reversal Curse, arXiv:2309.12288.
[2] Dziri et al. 2023, Faith and Fate, arXiv:2305.18654.
[3] Gloeckle et al. 2024, Multi-token Prediction, arXiv:2404.19737.
[4] LeCun 2022 AMI; Assran et al. 2025, V-JEPA2, arXiv:2506.09985.
[5] Hafner et al. 2023, DreamerV3, arXiv:2301.04104.
[6] Adhikari & Lapata 2026, Pearl, arXiv:2604.08065.
[7] Antoniou et al., GATE/EEVEE, OpenReview:0SMhqvgHST; TALI, https://huggingface.co/datasets/Antreas/TALI.
[8] Behrouz et al. 2025, Titans, arXiv:2501.00663.
[9] Wang et al. 2026, HRM-Text, arXiv:2605.20613.
[10] Wang et al. 2025, HRM, arXiv:2506.21734.
[11] Jolicoeur-Martineau 2025, TRM, arXiv:2510.04871.
[12] Antoniou 2026, MNEME validation, internal.
[13] Gu & Dao 2023, Mamba, arXiv:2312.00752.
[14] Raposo et al. 2024, Mixture-of-Depths, arXiv:2404.02258.
[15] Ericsson, Antoniou et al. 2024, einspace, arXiv:2405.20838.
[16] Aler Tubella et al. 2019, Glass-Box Gov
[17] EU AI Act, Reg.2024/1689.
[18] Lieder & Griffiths 2020, Resource Rationality

CAPABILITY GAP ADDRESSED 1,000

Explain how the solution creates new functionality (efficiency, reasoning, controllability, new use-cases) rather than just copying existing transformer models.

Larger transformers cannot close this gap by scale: they are stateless, token-bound and compute-uniform, so memory, planning, governance and action are bolted on after training. We make them native.

Efficiency: computation routes to informative states, not spread across everything; we target 100B-class behaviour from a ~1B model where the architecture earns it.

Reasoning: the model reasons in latent, multimodal state, simulating futures and comparing actions. Our memory-and-recursion experiments already show added capability a parameter-matched baseline cannot reach.

Controllability: persistent, editable memory lets knowledge be updated, forgotten or policy-patched without retraining; a governance layer traces every decision against explicit norms, making compliance inspectable.

New use-case: local, auditable intelligence for regulated and embodied systems, where remote compute is infeasible and compliance mandatory.

EXISTING ARTIFACTS 2,000

What prototypes, code bases, data pipelines, or models are already working? Attach links or short demos if possible.

Foundations (cross-cutting, peer-reviewed/data):

- MAML++ (ICLR 2019): meta-learning, the basis our self-improvement and evolutionary work build on; thousands of citations.
- einspace (NeurIPS 2024): compositional architecture search by evolution; foundation for C7.
- GATE (Antoniou & Storkey 2024, OpenReview): multi-modal/task/domain evaluation; our Stage-1 evaluation harness.
- TALI (HuggingFace 2023): temporally-aligned quadra-modal dataset (video/audio/text/image); grounds C2 world-modelling.
- TSP / Information Thermodynamics (Axiotic TR 2026): ablation studies grounding C4 compute-allocation. github.com/axiotic-ai/tsp

Components, each with working code (github.com/axiotic-ai):

- **C1** chronos (TRL 4): triad past/present/future-objective training; ablations at 500M parameters.
- **C2** gestalt (TRL 3): latent world-model decoder with multimodal early fusion.
- **C3** mneme (TRL 4): slot memory with learnable read/write heads; validated long-context recall (report in repo). anubis (TRL 3): recursive latent reasoning.
- **C4** soma (TRL 4): information-thermodynamic routing with live compute budget. deep-self-attention: multi-resolution attention.
- **C5** consensus (TRL 4): multi-agent error-correction multi-agent verification for closed-loop training-data filtering (research-archivum/multi-agent-orchestra).
- **C7** apex (TRL 4): evolutionary optimisation approaching backprop efficiency.
- Infra: forge / odysseus: Kueue scheduling, queue intelligence, multi-agent error-correction-verified pipelines.

Deployed:

- Symphonia (C5 technique): multi-perspective analyst consensus through structured deliberation; the machinery behind our self-improvement loop, deployed with NIHR for UK Government policy. arc-yh.nihr.ac.uk/research/projects/symphonia-llm-assisted-expert-consensus-platform
- TALOS (C6): neuro-symbolic policy circuit with audit-trace instrumentation; internally deployed governance.

TECHNOLOGY READINESS LEVEL (TRL) ASSESSMENT 1,000

Rate the maturity of your overall system and each major sub-component.

Integrated system: TRL 2-3 — all seven components have prototypes; the composition is not yet trained end-to-end. Stage 1 targets TRL 4: each component evaluated against a compute-matched 70-100B baseline, and any showing no measurable gain is dropped from Stage 2 unless it supplies needed capability.

Per component: C1 triad temporal objective: TRL 4, cross-validated at 500M. C2 multimodal world model: TRL 2-3, latent decoder, early fusion. C3 memory and latent reasoning: TRL 3-4, recall validated, ablations running. C4 compute-aware attention: TRL 3-4, routing ablations running. C5 self-improvement: TRL 5 as deployed Symphonia technique, TRL 3 as training filter. C6 built-in norms: TRL 2-3 prototype; TALOS internally deployed. C7 learning beyond backprop: TRL 4, evolution approaching backprop efficiency; einspace.

Foundations: Information Thermodynamics study (TRL 4) grounds C4. Training infrastructure (scheduling, queue, multi-agent error-correction-verified pipeline): TRL 5.

OPEN RESEARCH RISKS 1,000

Identify the parts that are still research-grade.

High risk, high reward by design: a falsified hypothesis still advances the field. The seven components sit at different interfaces and do not compete for the same capability; the open question is how they couple.

Still research-grade (TRL 2-3): the world-model decoder (C2), recursive latent reasoning (C3) and architecture-level norms (C6); the rest are TRL 4 prototypes.

Open risks and mitigations:

- Per-component gain: each adds a capability but may not lift the combined model → leave-one-out ablations.
- Integrability: objectives were designed not to conflict, but are untested composed → integration sprints, coupling probes.
- Gradient degradation (C2-C4): expected optimisation problems, met with evolution or better gradient flow.
- Self-improvement bias (C5): closed-loop training can amplify errors → multi-agent verification across iterations.
- Scale-dependent synergy: couplings found at 500M may not survive at 1B; we catalogue both outcomes, frontier knowledge either way.

COMPUTE REQUIREMENTS 1,000

List compute required for Stage 1, include total hours needed, GPU type, source and justification.

Total: ~225,000 H100-equivalent GPU-hours over 7 months.

Budget: EUR 1.015M (EUR 145k/month × 7), at a blended ~EUR 4.5 per H100-equivalent hour inclusive of storage, networking and orchestration (RunPod H100 $2.99/h, B200 $5.49/h; Nebius H200 $2.30/h committed).

GPU type and source: reserved H100 / H200 / B200 cloud capacity, with Axiotic-owned and partner nodes as upside.

Allocation:

- 65k hours: component ablations and baseline replication.
- 55k hours: pairwise integrations and negative-result sweeps.
- 60k hours: sub-1B multimodal and world-model prototypes.
- 45k hours: evaluation, red-team reruns, reproducibility.

Justification: Stage 1 is not the final 1B training run; it identifies which mechanisms earn Stage 2 scaling. Fixed baselines, early-stop gates, leave-one-component-out ablations and capability-per-watt accounting prevent us buying signal with brute force.

KPIS, BENCHMARKS AND POTENTIAL IMPACT

1,000

Which KPIs or benchmarks (if existing) would you use to track your progress? In what existing or future industry do you expect the highest impact? How would you assess it?

KPIs (Stage targets vs matched baselines):

- **Capability-per-Watt:** ≥10× at S1, ≥100× at S3.
- **Long-context:** ≥50% less degradation at 10× context (RULER).
- **Compositionality:** ≥20% over single-component (GSM-Symbolic).
- **Safety:** ≥90% (AgentDojo).
- **Auditability:** every inference traced.

Highest impact: embodied AI and robotics (a 1B action model fits on a robot, vehicle or factory cell), then regulated healthcare, defence and finance needing on-prem deployment, editable memory (GDPR) and audit governance (EU AI Act); German automotive and Nordic manufacturing are early markets.

Assessment: matched-FLOP runs vs SOTA, third-party reproductions, audit traces on deployed systems.

We are not promising magic. We are building the experiment that tells Europe whether structure can beat scale. If it works, it becomes a new model class. If some components fail, the Atlas still tells the field which paths were mirages. Either way, SPRIND buys frontier knowledge, not benchmark smoke.

WORK PLAN 4,000

The plan should:

- Cover milestones for the entire duration of the three stages, with specific information on what will be achieved after Stage 1.
- Indicate where collaboration and subcontracting work packages will be required.
- Contain cost information, including contributions by the team or the team's institution (e.g. personnel, licenses, infrastructure) for three stages of the Challenge.

Four WPs across 24 months. Stage 1 (M0–M7, EUR 3M, ~14 FTE): validate seven components + ship the Atlas. Stage 2 (M7–M15, EUR 8M, ~15 FTE): compose survivors into a 500M unified model. Stage 3 (M15–M24, EUR 15.5M, ~17 FTE): scale to 1B and ship.

WP1 — Project Management & Governance (M0–M24, lead: Dr Antoniou).

T1.1 PM: scrum cadence; exec committee (Antoniou chair, Bernal PM, Dr Theodorou governance).

T1.2 Advisory: Dr Yang (Meta), Dr Minervini, comp-neuro consultants (Dr Kaiser, Roy, Hajizada).

T1.3 Data: FAIR plan, licensing audit, eval-suite custody.

T1.4 Ethics: EU AI Act gates, RUS sanctions, IP register, Theodorou-chaired review.

WP2 — Dissemination & Exploitation (M0–M24, lead: Bernal).

T2.1 Dissemination: papers (ICML, NeurIPS, ICLR); Atlas v1 public (M7); eval dashboard; open repositories.

T2.2 Website: axiotic.ai pages, Symphonia case study, Atlas portal.

T2.3 Commercial: pilots with NHS (via Symphonia), Bayes Centre, Edinburgh ecosystem.

T2.4 VC: Series-A readiness, EUR 1B-class follow-on by Stage 3.

WP3 — Components R&D (M0–M24, lead: Dr Antoniou). S1: validate each sub-500M. S2: refine survivors at 500M. S3: scale to 1B.

T3.1 C1 temporal triad (Antoniou/Scannell/Charalampous): 10.6M model shows +79.7% phrase-fill (3 seeds); S1 cross-val at 500M / 5–10B tokens.

T3.2 C2 world model (Goel/Scannell): VLM grounding on GATE/TALI; S1 training-objective transfer to VLM.

T3.3 C3 memory + latent reasoning (Mai/Jones/McGimsey): MNEME recall validated; S1 ≥50% degradation cut at 32–128k vs Llama-3.

T3.4 C4 compute-aware attention (Mai/Barker, Yang adv.): routing baselines built, matching/beating N^2 attention on toy tasks at matched compute; S1 matched-FLOP parity vs dense ViT, ImageNet-1k.

T3.5 C5 self-improvement (Speggiorin/McGimsey/Charalampous): multi-agent error-correction judge on ICLR-25 papers (rho=0.74); S1 GAIA + SWE-Bench-Pro live.

T3.6 C6 norms (Dr Theodorou/Charalampous/Jones): EU AI Act, NIST, ISO 42001; S1 100% of inferences traced.

T3.7 C7 beyond backprop (Woehrle/Antoniou/Rezk): evolution at ~75% of an AdamW-trained model; S1 match AdamW on ResNet20 + 250M ES prototype.

WP4 — Integration & Validation (M4–M24, lead: Antoniou/Woehrle/Mai/Goel).

T4.1 Compositional research → Atlas: ~6,448 crossover runs (13 architectures × 4 tasks × 4 modalities) already map interactions; S1 pairwise coupling sweep + Atlas v1. The Atlas is a public, living record of every coupling we try, what worked and especially what did not, with approach, result and failure mode, so others avoid dead ends and retrace promising paths without repeating our mistakes.

T4.2 Validation: per-component reports vs compute-matched baselines, leave-one-out ablations, safety checks, audit-trace bridge.

Milestones. M7 (S1): Atlas v1 public; seven components validated vs compute-matched 70–100B baselines; gates met. M15 (S2): 500M composed model; first end-to-end C6 flywheel; HW Robotarium begins; hire edge MLOps. M24 (S3): 1B audited matched-FLOP vs 100B baselines (AgentBench, HELM, GAIA); production C6 flywheel; hire embodied lead.

Gates (public, fail-fast): K1≥2× / K2≥50% / K3≥5% / K6=100%. No component advances without a matched-FLOP audit; one with no metric gain but needed capability (e.g. governance) is retained.

Cost. S1 EUR 3M: compute 1.015M / personnel 1.460M / subcontracts 350k / ops 175k. S2 EUR 8M and S3 EUR 15.5M: team and compute scale ~3× per stage. In-kind: shipped repositories, 1,300+ tests, MTEB demonstrator with live revenue.

Collaboration & subcontracting. Dr Theodorou's institute (C6); Dr Mai/Edinburgh (C3–C4); Dr Goel/NVIDIA (C2); Speggiorin/Glasgow (C5). Shared labs: HW Robotarium (S2), Edinburgh Centre for Robotics (S3).

Long-term vision. Years 2–5: virtually embodied agents, a body and a mind per task. Years 5–10: into the physical world, construction to household to medicine. Ultimately, the ability to prompt atoms.

FINANCIAL COST ESTIMATE FOR STAGE 1 100

Max 3,000,000 EUR, in EUR.

3,000,000 EUR: compute 1.04m, personnel 1.40m, subcontracts 130k, ops 279k, contingency 151k

TEAM 2,000

Address:
• Why is your team best suited to achieve the goal of the challenge?
• What relevant field-specific expertise and experience do you already have, and with whom?
• What expertise is the team missing, and how will you address it?
• Attach the CVs of the key people in the Attachments section.

Axiotic is an interdisciplinary team across theoretical research, engineering, governance, commercialisation and business development, anchored in Edinburgh's ML ecosystem.

Founders (full-time):

- **Dr Antreas Antoniou** owns the compositional thesis: 8 years building the techniques the programme composes, and the only person combining meta-learning, multimodal and architecture search at this depth to architect the integration.
- **Samuel Jones** turns research into product: ex-Chief AI Officer at Pieces, where he shipped Long-Term Memory (LTM-2, #1 on Product Hunt) and solved memory-contamination in production, a direct C3 analogue.
- **Laura Bernal** drives fundraising: built Edinburgh's deep-tech venture builder (70+ startups, GBP 8M raised); Chief of Staff at Malted AI (GBP 7M round).
- **Dr Andreas Theodorou** leads AI governance: co-author of IEEE P7001, PI/participant in European projects exceeding EUR 25M.

Full-time engineers/researchers: Aidan Scannell (C2/C3, discrete-codebook world models), Kieran McGimsey (C4; shipped Pieces LTM with Jones), Alessandro Speggiorin (C6 judge infrastructure), Aristotelis Charalampous (C7/C1 encoders), Thomas Woehrle (C5 lead, evolution + architecture search).

Part-time specialists, mapped to prior work: Dr Chenhongyi Yang (Meta, C4 group-token attention), Dr Luo Mai (Edinburgh, C3/C4 routing), Dr Arushi Goel (NVIDIA, C2 vision-language), Dr Pasquale Minervini (advisor, C3 neuro-symbolic), Fady Rezk (C7/C4 engineering), Cameron Barker (consumer-GPU sparsity). Neuroscience-ML consultants: Dr Marcus Kaiser, Om Roy, Elvin Hajizada. 23K+ combined citations; peak h-index ~24 (Minervini).

Communications & growth: Hanna Stechenko (digital reach and the Atlas portal; ex-Pieces with Jones and McGimsey, scaled traffic 100K to 1M at PandaDoc).

Missing and fill: RL depth, edge MLOps, an embodied-integration lead — staged Stage-2/3 hires via Edinburgh CDT, Heriot-Watt Robotarium and the Edinburgh Centre for Robotics; no in-house robotics needed.

TEAM MEMBER 1-3 250

Name, role on the project and percent FTE committed

Antreas Antoniou,Cofounder, Chief Science Officer / Head of Research. DAEDALUS programme architect; owns C1, Compositional Research; supports C7 and Ogma. PhD Edinburgh (Storkey); MAML++, DAGAN, einspace NeurIPS 24. FTE: 100%.

TEAM MEMBER 1 links

Links to demonstrate track record (Industry, Entrepreneurship, Google-Scholar, GitHub-Project)

Site antreas.io · Scholar scholar.google.com/citations?user=LNO4XZQAAAAJ (7,972 cites) · GitHub github.com/AntreasAntoniou · LinkedIn linkedin.com/in/antreasantoniou · Axiotic axiotic.ai · ex-Pieces, Malted AI, Edinburgh postdoc, Google AI.

TEAM MEMBER 2 250

Laura Bernal Vergara, Cofounder, Head of Commercial. Owns partnerships, commercial validation, Series-A readiness, Edinburgh-ecosystem access (Bayes Centre, CRH, Barclays Eagle Labs). Ex-Chief of Staff Malted AI (£7M); built Edinburgh VBI. FTE: 100%.

TEAM MEMBER 2 links

LinkedIn linkedin.com/in/laurabernalvergara · Edinburgh VBI edinburgh-innovations.ed.ac.uk/vbi (70+ deep-tech startups, £8M raised in cohort) · Aspen Rising Leaders Fellow · Scottish AI Summit speaker · Axiotic axiotic.ai · ex-Malted AI CoS.

TEAM MEMBER 3

Samuel Jones, Cofounder, Head of Engineering; owns MNEME and the production demonstrator. Ex-CAIO at Pieces: built on-device long-term memory and edge models for 150k+ developers from scratch; scaled to 50, raised $27M. FTE: 100%

TEAM MEMBER 3 Links

LinkedIn: linkedin.com/in/samuel-jonathan-jones · Axiotic axiotic.ai · GitHub: github.com/sam-at-axiotic · ex-CAIO Pieces (on-device long-term memory for 150k+ developers); MSc AI Edinburgh, Distinction.

TEAM MEMBER 1-3 250

Name, role on the project and percent FTE committed

Antreas Antoniou,Cofounder, Chief Science Officer / Head of Research. DAEDALUS programme architect; owns CHRONOS, Compositional Research; supports APEX and Ogma. PhD Edinburgh (Storkey); MAML++, DAGAN, einspace NeurIPS 24. FTE: 100%.

TEAM MEMBER 1 links

Links to demonstrate track record (Industry, Entrepreneurship, Google-Scholar, GitHub-Project)

Site antreas.io · Scholar scholar.google.com/citations?user=LNO4XZQAAAAJ (7,972 cites) · GitHub github.com/AntreasAntoniou · LinkedIn linkedin.com/in/antreasantoniou · Axiotic axiotic.ai · ex-Pieces, Malted AI, Edinburgh postdoc, Google AI.

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TEAM MEMBER 3 Links

LinkedIn linkedin.com/in/samuel-jonathan-jones · GitHub github.com/sam-at-axiotic

ATTACHMENTS

Mandatory uploads:
• CVs of key personnel
• Detailed cost overview (PDF)
• Declaration on RUS Sanctions form (obligatory) – available in German and English on the SPRIND site
File format: .PDF

V2

TEAM

The four cofounders are the four people needed to ship the bet. Eleven founding team members lead the literature in their components; ~19,450 combined citations across 100+ articles (peak h-index 37 / i10 65, Ponti).

Co-founders

- **Antoniou (Head of Research).** The compositional thesis is his. Eight years on the techniques the model composes: MAML++ (ICLR'19), DAGAN, einspace (NeurIPS'24), TALI/GATE. Architects the integration.

- **Jones (Head of Engineering).** A decade of leadership experience across industries including 5 years in AI. Led orgs from 5 to 60 people. Raised $27 million in VC. Architected and shipped locally executing agentic workflows to hundreds of thousands of end users. Turns research into product at pace.
- **Bernal (Head of Commercial).** Built Edinburgh's Venture Builder Initiative (70+ startups, £8M raised); CoS at Malted AI through its £7M round. Drives Series-A.
- **Theodorou (Head of AI Governance).** Co-authored IEEE P7001 — the transparency standard regulators read. Audit traces are co-designed with the architecture.

Founding Team (Edinburgh, Heriot-Watt, Glasgow, Meta, AMAZON, EY). Each leads their component.

- Mai (Edinburgh): WaferLLM / ServerlessLLM — scales compute routing past single-GPU.
- Ponti (Edinburgh): Dynamic Memory Compression, Library-of-LoRAs — memory foundation.
- Suglia (Heriot-Watt): AlanaVLM (beat GPT-4V on OpenEQA) — multi-modal grounding.
- Scannell (Edinburgh): Discrete Codebook World Models (ICLR'25) — world-model rollouts.
- Yang (Meta): GPViT group tokens — relational binding precedent.
- Speggiorin (Glasgow): TaskMAD / COMEX — judge infrastructure.
- Barker (Edinburgh PhD): sparsity, quantisation, consumer-GPU engineering.
- Woehrle (Edinburgh PhD): Natural Evolution Strategies + einspace — evolutionary optimisation lead.
- Charalampous (EY): HKAN encoders — audit traces and triad-temporal training.
- McGimsey (Axiotic): relational reasoning and production demonstrator — shipping.

Gaps: RL beyond Scannell → Stage 2 Edinburgh CDT hire. Edge MLOps → Stage 2 or Heriot-Watt Robotarium.

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Site antreas.io · Scholar scholar.google.com/citations?user=LNO4XZQAAAAJ (7,972 cites) · GitHub github.com/AntreasAntoniou · LinkedIn linkedin.com/in/antreasantoniou · Axiotic axiotic.ai · ex-Pieces, Malted AI, Edinburgh postdoc, Google AI.

TEAM MEMBER 2 250

Laura Bernal Vergara,Cofounder, Head of Commercial. Owns partnerships, commercial validation, Series-A readiness, Edinburgh-ecosystem access (Bayes Centre, CRH, Barclays Eagle Labs). Ex-Chief of Staff Malted AI (£7M); built Edinburgh VBI. FTE: 100%.

TEAM MEMBER 2 links

LinkedIn linkedin.com/in/laurabernalvergara · Edinburgh VBI edinburgh-innovations.ed.ac.uk/vbi (70+ deep-tech startups, £8M raised in cohort) · Aspen Rising Leaders Fellow · Scottish AI Summit speaker · Axiotic axiotic.ai · ex-Malted AI CoS.

TEAM MEMBER 3

Samuel Jones, Cofounder, Head of Engineering; owns MNEME and the production demonstrator. Ex-CAIO at Pieces: built on-device long-term memory and edge models for 150k+ developers from scratch; scaled to 50, raised $27M. MSc AI, Distinction. FTE: 100%

TEAM MEMBER 3 Links

LinkedIn linkedin.com/in/samuel-jonathan-jones · GitHub github.com/sam-at-axiotic

ATTACHMENTS

Mandatory uploads:

• CVs of key personnel
• Detailed cost overview (PDF)
• Declaration on RUS Sanctions form (obligatory) – available in German and English on the SPRIND site

File format: .PDF

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